

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: INJONGO | Context: Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

HIGH-priority dimension with substantial concerns. INJONGO followed a rigorous quality control process (three annotators per utterance [\[Q42, Q45\]](#), two review rounds [\[Q109\]](#), post-review Kappa 0.912–1.00 [\[Q48\]](#)). However, for the Ethiopian utility deployment specifically: (1) Ground-truth authority mismatch — annotators were recruited via a Kenya-based logistics company at per-country rates [\[Q44\]](#), and there is no documentation that Amharic annotators had EEU/AAWSA utility billing domain expertise (gap_id 3 NOT FOUND); the deployment elicitation explicitly requires utility agent validation. (2) Annotation quality issues observable in samples — D15 and D16 show Amharic utterances with clearly extractable DATE, TIME, and CALENDAR_EVENT entities but empty span annotations; D13 (Oromo) and D21 (Sesotho) show similar omissions. (3) Amharic-specific Fleiss Kappa not surfaced (gap_id 7 NOT FOUND), so where Amharic falls in the 0.618–0.934 pre-review range is unknown. (4) Pre-to-post improvements were largest in Sesotho and Zulu [\[Q49\]](#), suggesting language-specific quality variation that is not transparent for Amharic.

ID	Evidence
Q42	"Each utterance was annotated by three annotators so that we could check for agreement in the slot annotations." (p.4)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q45	"To ensure annotation quality and consistency, we follow a rigorous quality control process using a majority voting system with a minimum of three annotators per sentence to resolve disagreements." (p.4)
Q48	"Following the review process, agreement scores improved markedly across all languages, reaching 0.912-1.00." (p.4)
Q49	"Notable improvements were observed in Sesotho (+0.327) and Zulu (+0.294), with other languages showing average improvements of approximately 0.1 in their Fleiss' Kappa scores." (p.4)
Q109	"The annotation process revealed inherent challenges in entity classification across languages, requiring two rounds of review to achieve consistent quality." (p.9)
D13	"Balallii Dirree dawaa irraa gara Boolee achiin gara jimmaa Xiyyaara Itoopiyaa irraa naaf kuti" — no spans — Oromo: "Book me a flight from Dire Dawa to Bole then to Jimma on Ethiopian Airlines" — Bole airport and Dire Dawa named, spans incorrectly empty
D15	"ነገ ጠዋት 12 ሰዓት የቀጠርኩትን ማንቂያ በደንብ እንዲሰማኝ ድምጽ ጨምርበት" — spans: [] — "Increase the volume of the alarm I set for 12 o'clock tomorrow morning so I can hear it well." — TIME and DATE slots present in text but no spans annotated
D16	"የእናቴን እና የአባቴን የልደት ቀን የቀን መቁጠሪያዬ ላይ ስላቀያየርኩት አስተካክለዋል" — spans: [] — "Fix it because I changed my mother's and father's birthday on my calendar." — CALENDAR_EVENT-like content but no spans annotated
D21	"Ke hloka bokae ho lefa karete ya Edgars" — spans: [] — Sesotho: "How much do I need to pay the Edgars card?" — Edgars is South African retail credit card; no slots annotated despite implied BILL_TYPE/SHOPPING_ITEM

Strengths

- Three-annotator majority-vote QC with two-round review [\[Q42, Q45, Q109\]](#)
- Post-review Fleiss Kappa 0.912–1.00 across all languages [\[Q48\]](#)
- Detailed annotation guide with explicit disambiguation rules [\[Q149, Q152, Q153, Q154, Q155\]](#)
- Native-speaker annotators across all languages [\[Q7\]](#)
- Per-country remuneration via professional logistics company [\[Q44\]](#)

ID	Evidence
Q7	"utterances generated by native speakers across diverse domains, including banking, travel, home, and dining." (p.1)
Q42	"Each utterance was annotated by three annotators so that we could check for agreement in the slot annotations." (p.4)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
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Q48	"Following the review process, agreement scores improved markedly across all languages, reaching 0.912-1.00." (p.4)
Q109	"The annotation process revealed inherent challenges in entity classification across languages, requiring two rounds of review to achieve consistent quality." (p.9)
Q149	"This section provides the complete annotation guide and instruction for annotators working for labeling all slots types." (p.20)
Q152	"When annotating Bank names, you do not need to include "bank" unless it is attached to the bank name, like seen above, with Ecobank." (p.21)
Q153	"Anything that is less than one day should be annotated as TIME and not DATE, as seen in the above examples." (p.21)
Q154	"Anything that is more than one day must be annotated as DATE and not time, as seen above" (p.21)
Q155	"You include "bill" as part of the annotation." (p.21)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — ground truth was set by pan-African native-speaker crowdworkers, not by EEU/AAWSA customer service agents as the deployment requires. The deployment elicitation explicitly states utility agent/billing staff judgment determines correctness.

OC-2: Likely substantial for utility-specific spans (BILL_TYPE boundaries, place-name resolution to kebele/sub-city). D7 ('የቦሎ ማሳደሻ' annotated as BILL_TYPE) — whether this matches EEU/AAWSA categorization cannot be confirmed without domain-expert review. Cultural conventions around formal Amharic register also vary by service institution.

OC-3: Demographics partially documented: native speakers via Kenya-based logistics company [Q44]; per-language counts and Ethiopia-specific recruitment unspecified (NOT FOUND in YAML benchmark_amharic_split_details). No domain-expertise requirement documented.

OC-4: Re-annotation by EEU and AAWSA staff (or supervised review) is needed for any utility-specific slot labels before deployment use. The deployment explicitly requires this.

OC-5: Aggregation method is majority vote with review [Q45]; minority-perspective erasure risk is moderate, particularly for ambiguous slot types where annotators disagreed [Q53]. Reviewer identity and resolution rules are not detailed.

OC-6: Convergent validity concern: Amharic ground-truth labels may not correlate with EEU/AAWSA agent judgments. External validity concern: under-annotation observed in samples [D15, D16] reduces reliability of slot-filling F1 as a deployment-readiness signal.

ID	Evidence
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Q7	"utterances generated by native speakers across diverse domains, including banking, travel, home, and dining." (p.1)
Q44	"For both utterance elicitation and slot-filling annotation, all recruited participants received an appropriate remuneration based on the per-country rate decided by our logistics company in Kenya." (p.4)
Q45	"To ensure annotation quality and consistency, we follow a rigorous quality control process using a majority voting system with a minimum of three annotators per sentence to resolve disagreements." (p.4)
Q53	"Examination of annotator feedback and comparative analysis between unreviewed and reviewed versions indicated that ambiguous slot types significantly impacted annotation quality and introduced unnecessary complexity." (p.4)
D7	"የቦሌ ማሳደሻ ስንት ነው መክፈል የሚኖርብኝ?" — BILL_TYPE: የቦሌ ማሳደሻ — "How much do I need to pay for Bole road expansion?" — bill query referencing Bole (Addis Ababa neighborhood)
D15	"ነገ ጠዋት 12 ሰዓት የቀጠርኩትን ማንቁያ በደንብ እንዲሰማኝ ድምጽ ጨምርሰኝ" — spans: [] — "Increase the volume of the alarm I set for 12 o'clock tomorrow morning so I can hear it well." — TIME and DATE slots present in text but no spans annotated
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WEB-9	INJONGO paper — gap_id 3 NOT FOUND on Amharic annotator domain expertise; gap_id 7 NOT FOUND on Amharic-specific Kappa (https://arxiv.org/abs/2502.09814)

Information Gaps

- Amharic-specific pre- and post-review Fleiss Kappa values (gap_id 7)
- Per-language annotator counts and locations for Amharic (NOT FOUND)
- Whether reviewers had domain expertise in any specific industry

Requires Expert Verification

- Whether EEU/AAWSA agents agree with sampled BILL_TYPE boundary decisions in Amharic data
- Whether under-annotated examples [D15, D16] are isolated or systematic across the full Amharic split

Remediation

Gap 1: Annotators were pan-African native-speaker crowdworkers with no documented utility domain expertise; deployment requires EEU/AAWSA agent validation as ground-truth authority.

Recommendation: Have EEU and AAWSA customer service agents re-validate a stratified sample of the Amharic test split's utility-relevant labels; resolve discrepancies; retain only utility-domain examples whose labels match agent judgment for deployment-readiness evaluation.